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Saveetha Institute of Medical And Technical Sciences

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ITA0513 – COMPUTER VISION

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ASSIGNMENT – 3

Feature Detection and Matching

Course Code	Course Name	Assessment	CO	BTL	Max. Marks
ITA0513	Computer Vision	Assignment – 3	CO3	BTL4	50

CO3: Analyse and extract image features using detection and matching techniques for effective image representation and comparison.

Instructions: Answer all five questions (10 marks each). Show every computational step, state formulas before substituting values, and justify each decision using your computed results.

Mark Allocation

Q. No.	Topic	(a)	(b)	(c)	(d)	Total
1	Harris corner detection	3	3	2	2	10
2	Hough transform for lines	3	3	2	2	10
3	Principal component analysis	3	3	2	2	10
4	Scale-invariant feature matching (SIFT)	3	3	2	2	10
5	Boundary representation (chain codes)	3	3	2	2	10
	Total	15	15	10	10	50

Question 1 (CO3, BTL4, 10 Marks) – Harris Corner Analysis for Drone Visual Odometry

A delivery drone estimates its motion by tracking corners on rooftops. The Harris detector uses the structure tensor $M = \Sigma[I_x^2\ I_xI_y;\ I_xI_y\ I_y^2]$ over a 3 × 3 window with uniform weights and the response $R = \det(M) - k\cdot[\text{trace}(M)]^2$ with $k = 0.04$. A window is classified as a corner if $R > 10^6$, an edge if $R < 0$ and flat otherwise. The gradients of window W1 are given below; windows W2 and W3 have precomputed tensors $\Sigma(I_x^2, I_y^2, I_xI_y) = (4800, 4500, 300)$ and $(150, 140, 20)$.

Window W1	Row 1	Row 2	Row 3
I_x	12 15 10	14 18 11	13 16 12
I_y	11 9 14	13 17 10	12 15 13

(a) Compute ΣI_x^2, ΣI_y^2 and ΣI_xI_y for W1 and form the structure tensor M. [3 Marks]

(b) Compute det(M), trace(M), R and the eigenvalues of M for W1, and classify the window. [3 Marks]

(c) Compute R for W2 and W3, classify them, and rank W1, W2 and W3 by suitability for tracking. [2 Marks]

(d) Analyse why the Harris response is invariant to image rotation but not to scale, and the consequence for a drone that changes altitude. [2 Marks]

ANSWER:

(a) Calculate ΣIx², ΣIy² and ΣIxIy for W1 and form the structure tensor

Given:

$$I_x= \begin{bmatrix} 12&15&10\\ 14&18&11\\ 13&16&12 \end{bmatrix}$$

$$I_y= \begin{bmatrix} 11&9&14\\ 13&17&10\\ 12&15&13 \end{bmatrix}$$

Step 1: Calculate ΣIx^2

$$\sum I_x^2=12^2+15^2+10^2+\cdots+12^2$$

$$=144+225+100+196+324+121+169+256+144$$

$$\boxed{\sum I_x^2=1679}$$

Step 2: Calculate ΣIy^2

$$\sum I_y^2=11^2+9^2+14^2+\cdots+13^2$$

$$\boxed{\sum I_y^2=1494}$$

Step 3: Calculate $\Sigma IxIy$

$$\sum I_xI_y=(12\times11)+(15\times9)+\cdots+(12\times13)$$

$$\boxed{\sum I_xI_y=1557}$$

Structure Tensor:

$$ M= \begin{bmatrix} \sum I_x^2&\sum I_xI_y\\ \sum I_xI_y&\sum I_y^2 \end{bmatrix} $$

$$ \boxed{ M= \begin{bmatrix} 1679&1557\\ 1557&1494 \end{bmatrix} } $$

(b) Calculate det(M), trace(M), R and eigenvalues. Classify W1.

Step 1: Determinant

$$ det(M)=(1679\times1494)-(1557)^2 $$

$$ =2508426-2424249 $$

$$ \boxed{det(M)=84177} $$

Step 2: Trace

$$ trace(M)=1679+1494 $$

$$ \boxed{trace(M)=3173} $$

Step 3: Harris Response

Formula:

$$ R=det(M)-k[trace(M)]^2 $$

Given $k=0.04$.

$$ R=84177-0.04(3173)^2 $$

$$ R=84177-402717.16 $$

$$ \boxed{R=-318540.16} $$

Step 4: Eigenvalues

Using the eigenvalue formula:

$$ \lambda=\frac{(a+d)\pm\sqrt{(a-d)^2+4b^2}}{2} $$

Substituting the values:

$$ \lambda_1\approx3146.25 $$

$$ \lambda_2\approx26.75 $$

Classification:

Since R is negative:

$$ \boxed{W1=\text{Edge}} $$

(c) Calculate R for W2 and W3, classify and rank them.

For W2:

Given:

$$ \sum I_x^2=4800,\quad \sum I_y^2=4500 $$

$$ \sum I_xI_y=300 $$

$$ det(M)=(4800\times4500)-300^2 $$

$$ =21510000 $$

$$ trace(M)=4800+4500=9300 $$

$$ R=21510000-0.04(9300)^2 $$

$$ \boxed{R=18050400} $$

W2 – Corner

For W3:

Given:

$$ \sum I_x^2=150,\quad \sum I_y^2=140 $$

$$ \sum I_xI_y=20 $$

$$ det(M)=(150\times140)-20^2 $$

$$ =20600 $$

$$ trace(M)=150+140=290 $$

$$ R=20600-0.04(290)^2 $$

$$ \boxed{R=17236} $$

W3 – Flat

Assuming the given threshold 10^6.

Final Classification and Ranking:

Window	Harris Response (R)	Classification
W1	-318540.16	Edge
W2	18050400	Corner
W3	17236	Flat

Tracking suitability ranking:

- W2 – Best: Corner features are suitable for accurate tracking in both directions.
- W1 – Moderate: Edge features provide information mainly in one direction.
- W3 – Poor: Flat regions contain very little useful gradient information.

(d) Explain rotation and scale invariance and the effect of drone altitude. [2 Marks]

Rotation Invariance:

- Harris corner detection is invariant to image rotation.
- When an image rotates, the eigenvalues, determinant and trace of the structure tensor remain unchanged.
- Therefore, the corner response remains the same.

Scale Sensitivity:

- Harris detector is not naturally scale invariant.
- When the drone changes altitude, the apparent size of rooftop objects changes.
- At higher altitude, objects appear smaller, and at lower altitude, they appear larger.
- This may affect corner detection and tracking accuracy.

Solution:

A scale pyramid or multi-scale Harris detector can be used to detect corners at different sizes and improve drone tracking accuracy.

Conclusion: The Harris detector identifies W2 as the best corner feature for drone visual odometry because it provides strong information in both directions.

Question 2 (CO3, BTL4, 10 Marks) – Hough Transform Analysis for Lane-Marking Detection

An advanced driver-assistance system detects lane markings from edge pixels in a bird's-eye-view image. A small test set of five edge points is given below. The normal-form Hough transform $\rho = x \cos\theta + y \sin\theta$ is evaluated at $\theta \in \{0°, 45°, 90°, 135°\}$, and ρ is quantised to the nearest integer when voting into the accumulator.

Point	**E1**	**E2**	**E3**	**E4**	**E5**
(x, y)	(2, 6)	(4, 4)	(6, 2)	(1, 7)	(3, 1)

(a) Compute ρ for every point at every θ, and present the results in a table. [3 Marks]

(b) Fill the accumulator after quantising ρ, and identify the peak. [3 Marks]

(c) Convert the peak to the line equation in image coordinates and identify the outlier. [2 Marks]

(d) Analyse the effect of the coarse θ resolution and of the ρ quantisation on this result, and explain why the Hough transform is preferred over least-squares line fitting here. [2 Marks]

ANSWER:

(a) Calculate ρ for every point at every θ.

Given normal-form Hough transform:

$$\rho=x\cos\theta+y\sin\theta$$

Angles:

$$\theta=0^\circ,45^\circ,90^\circ,135^\circ$$

Given points:

- E1 = (2, 6)
- E2 = (4, 4)
- E3 = (6, 2)
- E4 = (1, 7)
- E5 = (3, 1)

Step 1: Calculate ρ values

At 0°:

$$\rho=x$$

At 45°:

$$\rho=\frac{x+y}{\sqrt{2}}$$

At 90°:

$$\rho=y$$

At 135°:

$$\rho=\frac{-x+y}{\sqrt{2}}$$

Calculation table:

Point	0°	45°	90°	135°
E1 (2,6)	2	5.66	6	2.83
E2 (4,4)	4	5.66	4	0
E3 (6,2)	6	5.66	2	-2.83
E4 (1,7)	1	5.66	7	4.24
E5 (3,1)	3	2.83	1	-1.41

(b) Fill the accumulator and identify the peak.

The ρ values are rounded to the nearest integer.

Accumulator table:

θ	Quantised ρ values	Votes
0°	1, 2, 3, 4, 6	1 each
45°	6, 6, 6, 6, 3	ρ = 6 has 4 votes
90°	1, 2, 4, 6, 7	1 each
135°	-3, -1, 0, 3, 4	1 each

Peak detected

$$\boxed{\theta=45^\circ,\quad \rho=6}$$

The peak receives 4 votes, which indicates a strong line.

(c) Convert the peak into a line equation and identify the outlier.

Using the Hough normal equation:

$$\rho=x\cos\theta+y\sin\theta$$

Substitute $\rho=6$ and $\theta=45^\circ$:

$$6=x\cos45^\circ+y\sin45^\circ$$

$$6=\frac{x+y}{\sqrt{2}}$$

$$\boxed{x+y=8}$$

Check the points

Point	x + y	Result
E1	8	On line
E2	8	On line

E3	8	On line
E4	8	On line
E5	4	Outlier

Therefore,

$$\boxed{E5=(3,1)\text{ is the outlier}}$$

(d) Analyse θ resolution, ρ quantisation and least-squares fitting.

1. Effect of coarse θ resolution:

- Only four angles are considered.
- The actual lane direction may fall between these angles.
- This reduces detection accuracy.
- Using smaller angular steps improves accuracy.

2. Effect of ρ quantisation:

- Rounding ρ to the nearest integer reduces precision.
- Different points may vote for the same accumulator cell.
- This can cause small errors in line detection.

3. Why Hough Transform is preferred over least-squares fitting:

- Hough Transform can detect lines even when some edge points are noisy or outliers.
- It uses voting to identify the strongest line.
- Least-squares fitting is more sensitive to outliers.
- Therefore, Hough Transform is more suitable for detecting lane markings in noisy images.

Conclusion

The Hough Transform detects the lane line as $x + y = 8$ with four votes. E5 is identified as the outlier. It provides reliable line detection even when unwanted edge points are present.

Question 3 (CO3, BTL4, 10 Marks) – PCA for Dimensionality Reduction in Leaf-Disease Features

An agritech start-up classifies leaf diseases using two shape features per lesion: x_1 = lesion length (mm) and x_2 = lesion width (mm). To reduce storage and noise, the engineer analyses five training lesions with principal component analysis, using the sample covariance (divide by n − 1).

Lesion	L1	L2	L3	L4	L5
x_1	2	4	6	8	10
x_2	3	5	4	8	10

(a) Compute the mean vector and the sample covariance matrix. [3 Marks]

(b) Compute the eigenvalues and the principal eigenvector, and the percentage of variance explained by each component. [3 Marks]

(c) Project the five lesions onto PC1. [2 Marks]

(d) Analyse whether a single feature should replace the two original features, and interpret PC1 physically. [2 Marks]

ANSWER:

(a) Calculate the mean vector and sample covariance matrix.

Given data:

Lesion	x1 (Length)	x2 (Width)
L1	2	3
L2	4	5
L3	6	4
L4	8	8
L5	10	10

Step 1: Calculate mean of x1

$$\bar{x_1}=\frac{2+4+6+8+10}{5}$$

$$\boxed{\bar{x_1}=6}$$

Step 2: Calculate mean of x2

$$\bar{x_2}=\frac{3+5+4+8+10}{5}$$

$$\boxed{\bar{x_2}=6}$$

Mean vector:

$$\boxed{\mu=\begin{bmatrix}6\\6\end{bmatrix}}$$

Step 3: Calculate covariance matrix

Deviation table:

Lesion	x1 − 6	x2 − 6
L1	-4	-3
L2	-2	-1
L3	0	-2
L4	2	2
L5	4	4

Covariance formula:

$$ Cov(X)=\frac{1}{n-1} \begin{bmatrix} \sum(x_1-\bar{x_1})^2 & \sum(x_1-\bar{x_1})(x_2-\bar{x_2})\\ \sum(x_1-\bar{x_1})(x_2-\bar{x_2}) & \sum(x_2-\bar{x_2})^2 \end{bmatrix} $$

Calculations:

$$ \sum(x_1-\bar{x_1})^2=40 $$

$$ \sum(x_2-\bar{x_2})^2=34 $$

$$ \sum(x_1-\bar{x_1})(x_2-\bar{x_2})=34 $$

Since n = 5:

$$ Cov(X)=\frac{1}{4} \begin{bmatrix} 40&34\\ 34&34 \end{bmatrix} $$

$$ \boxed{ Cov(X)= \begin{bmatrix} 10&8.5\\ 8.5&8.5 \end{bmatrix} } $$

(b) Calculate eigenvalues, principal eigenvector and percentage of variance.

Step 1: Calculate eigenvalues

Using:

$$ |C-\lambda I|=0 $$

$$ \begin{vmatrix} 10-\lambda&8.5\\ 8.5&8.5-\lambda \end{vmatrix}=0 $$

$$ (10-\lambda)(8.5-\lambda)-8.5^2=0 $$

$$ \lambda^2-18.5\lambda+12.75=0 $$

Solving:

$$ \boxed{\lambda_1=17.783} $$

$$ \boxed{\lambda_2=0.717} $$

Step 2: Principal eigenvector

For the largest eigenvalue:

$$ \lambda_1=17.783 $$

The normalized principal eigenvector is:

$$ \boxed{ PC1= \begin{bmatrix} 0.738\\ 0.675 \end{bmatrix} } $$

(c) Project the five lesions onto PC1.

The projection formula is:

$$PC1=(x_1-6)(0.738)+(x_2-6)(0.675)$$

Projection calculations:

Lesion	PC1 projection
L1	-4.976
L2	-2.150
L3	-1.351
L4	2.826
L5	5.651

Therefore, the projected values are:

$$\boxed{[-4.976,-2.150,-1.351,2.826,5.651]}$$

Step 2: Percentage of variance explained

$$Variance\%=\frac{\lambda_i}{\lambda_1+\lambda_2}\times100$$

For PC1:

$$=\frac{17.783}{17.783+0.717}\times100$$

$$\boxed{96.12\%}$$

For PC2:

$$=\frac{0.717}{18.5}\times100$$

$$\boxed{3.88\%}$$

(d) Analyse whether a single feature should replace two features and interpret PC1 physically.

1. Should one feature replace two features?
   - PC1 explains 96.12% of the total variance.
   - PC2 explains only 3.88% of the variance.
   - Therefore, one principal component can replace the two original features with very little information loss.
   - This reduces storage and computational complexity.
2. Physical interpretation of PC1:
   - PC1 combines lesion length and width.
   - Both features have positive weights (0.738 and 0.675).
   - It represents the overall size of the leaf lesion.
   - Larger PC1 values indicate larger lesions.

Conclusion:

PCA successfully reduces two features into one principal component while retaining approximately 96.12% of the information. This helps improve storage efficiency and reduce noise in leaf-disease classification.

Question 4 (CO3, BTL4, 10 Marks) – SIFT Scale-Space and Descriptor Matching for Product Recognition

A retail app recognises packaged products on shelves by matching SIFT features against a product database. The detector uses a base scale $\sigma_0 = 1.6$ with S = 3 intervals per octave. For fast analysis, the 128-dimensional descriptors are represented here by 4-dimensional summaries. Three query descriptors from a shelf photo and three database descriptors are given. Lowe's ratio test accepts a match if $d_1/d_2 < 0.8$, where d_1 and d_2 are the nearest and second-nearest Euclidean distances.

Descriptor	**Values**
D1 (database)	14, 38, 9, 22
D2 (database)	30, 12, 20, 19
D3 (database)	11, 44, 4, 30
Q1 (query)	12, 40, 7, 25
Q2 (query)	30, 10, 22, 18
Q3 (query)	12.5, 41, 6.5, 26

(a) Compute $k = 2^{1/S}$, the scales of the S + 3 Gaussian images in the first octave, the number of DoG images, and the length of a full SIFT descriptor. [3 Marks]

(b) Compute the Euclidean distances from Q1, Q2 and Q3 to D1, D2 and D3. [3 Marks]

(c) Apply the ratio test to each query and state the accepted matches. [2 Marks]

(d) Analyse why the ratio test is more reliable than a fixed distance threshold, using Q3 as evidence, and why SIFT tolerates shelf photos taken at different distances. [2 Marks]

ANSWER:

SIFT Scale-Space and Descriptor Matching for Product Recognition

(a) Calculate k, Gaussian scales, DoG images and descriptor length.

Given:

- Base scale, \(\sigma_0=1.6\)

- Number of intervals, $S=3$

Step 1: Calculate k

$$ k=2^{1/S} $$

$$ k=2^{1/3} $$

$$ \boxed{k=1.2599} $$

Step 2: Calculate Gaussian image scales

Formula:

$$ \sigma_s=\sigma_0\times k^s $$

Since S = 3, the number of Gaussian images is:

$$ S+3=6 $$

Gaussian image	Scale
G0	1.600
G1	2.016
G2	2.540
G3	3.200
G4	4.032
G5	5.079

Step 3: Number of DoG images

$$ DoG=(S+3)-1 $$

$$ \boxed{DoG=5} $$

Step 4: SIFT descriptor length

A SIFT descriptor contains:

- 4 × 4 spatial regions
- 8 orientation bins per region

$$ 4\times4\times8=128 $$

$$ \boxed{\text{Descriptor length}=128} $$

(b) Calculate Euclidean distances between query and database descriptors. [3 Marks]

Formula:

$$ d=\sqrt{(q_1-d_1)^2+(q_2-d_2)^2+(q_3-d_3)^2+(q_4-d_4)^2} $$

Step 1: Q1 distances

Q1 = (12, 40, 7, 25)

Distance from D1:

$$ d(Q1,D1)=\sqrt{(-2)^2+2^2+(-2)^2+3^2} $$

$$ =\sqrt{21}=\boxed{4.583} $$

Distance from D2:

$$ d(Q1,D2)=\sqrt{(-18)^2+28^2+(-13)^2+6^2} $$

$$ =\boxed{36.235} $$

Distance from D3:

$$ d(Q1,D3)=\sqrt{1^2+(-4)^2+3^2+(-5)^2} $$

$$ =\boxed{7.141} $$

Step 2: Q2 distances

Q2 = (30, 10, 22, 18)

$$ d(Q2,D1)=\boxed{35} $$

$$ d(Q2,D2)=\boxed{3} $$

$$ d(Q2,D3)=\boxed{44.553} $$

Step 3: Q3 distances

Q3 = (12.5, 41, 6.5, 26)

$$ d(Q3,D1)=\boxed{5.788} $$

$$ d(Q3,D2)=\boxed{37.128} $$

$$ d(Q3,D3)=\boxed{5.788} $$

Final distance table:

Query	D1	D2	D3
Q1	4.583	36.235	7.141
Q2	35.000	3.000	44.553
Q3	5.788	37.128	5.788

(c) Apply Lowe's ratio test and identify accepted matches. [2 Marks]

Given:

$$ \frac{d_1}{d_2}<0.8 $$

Where:

- d_1 = nearest distance.
- d_2 = second-nearest distance.

For Q1:

Nearest = 4.583 (D1)

Second nearest = 7.141 (D3)

$$ Ratio=\frac{4.583}{7.141} $$

$$ \boxed{Ratio=0.641} $$

Since 0.641 is less than 0.8, Q1 is accepted and matched with D1.

For Q2:

Nearest = 3 (D2)

Second nearest = 35 (D1)

$$ Ratio=\frac{3}{35} $$

$$ \boxed{Ratio=0.086} $$

Since 0.086 is less than 0.8, Q2 is accepted and matched with D2.

For Q3:

Nearest = 5.788 (D1)

Second nearest = 5.788 (D3)

$$ Ratio=\frac{5.788}{5.788} $$

$$ \boxed{Ratio=1} $$

Since 1 is greater than 0.8, Q3 is rejected.

Final matching table:

Query	Best match	Ratio	Result
Q1	D1	0.641	Accepted
Q2	D2	0.086	Accepted
Q3	D1 / D3	1.000	Rejected

(d) Analyse the ratio test and explain SIFT scale tolerance. [2 Marks]

Why is the ratio test more reliable than a fixed distance threshold?

- A fixed distance threshold checks only the nearest distance.
- The ratio test compares the nearest and second-nearest distances.
- It avoids ambiguous matches.
- In Q3, both D1 and D3 have equal distances (5.788).
- Therefore, the ratio is 1, and the match is rejected.

Why does SIFT tolerate different camera distances?

- SIFT detects features at multiple scales using Gaussian scale-space.
- It identifies keypoints at different image sizes.
- It uses orientation information to handle image rotation.
- Its normalized descriptors help match products even when their apparent sizes change.

Conclusion:

SIFT provides reliable product recognition by detecting features at different scales and matching descriptors. Here, Q1 and Q2 are accepted, while Q3 is rejected because of ambiguity.

Question 5 (CO3, BTL4, 10 Marks) – Chain-Code Representation for Machined-Part Shape Inspection

A machine shop verifies the profile of punched sheet-metal parts using boundary descriptors that must be independent of the part's position and orientation on the conveyor. The traced boundary of a part, with x to the right and y upward, visits the grid points (0, 0) → (1, 0) → (2, 0) → (3, 1) → (3, 2) → (2, 3) → (1, 3) → (0, 2) → (0, 1) → (0, 0). The 8-direction Freeman code is used: 0 = east, 1 = north-east, 2 = north, 3 = north-west, 4 = west, 5 = south-west, 6 = south and 7 = south-east.

(a) Derive the 8-direction Freeman chain code of the boundary and normalise it with respect to the starting point. [3 Marks]

(b) Compute the circular first difference and the shape number. [3 Marks]

(c) Compute the perimeter from the chain code and the enclosed area of the polygon. [2 Marks]

(d) The part is rotated by 90° counter-clockwise on the conveyor. Derive the new chain code and first difference, and analyse which descriptor should be stored as the inspection template. [2 Marks]

ANSWER:

(a) Derive the 8-direction Freeman chain code and normalise it. [3 Marks]

Given boundary points:

(0,0) → (1,0) → (2,0) → (3,1) → (3,2) → (2,3) → (1,3) → (0,2) → (0,1) → (0,0)

Step 1: Identify the directions

Given Freeman direction codes:

Direction	Code
East	0
North-East	1
North	2
North-West	3
West	4
South-West	5
South	6
South-East	7

Step 2: Derive the chain code

Movement	Direction	Code
(0,0) → (1,0)	East	0

(1,0) → (2,0)	East	0
(2,0) → (3,1)	North-East	1
(3,1) → (3,2)	North	2
(3,2) → (2,3)	North-West	3
(2,3) → (1,3)	West	4
(1,3) → (0,2)	South-West	5
(0,2) → (0,1)	South	6
(0,1) → (0,0)	South	6

Correction: The movement from (1,3) to (0,2) is South-West, so its code is 5. The movement from (0,2) to (0,1) is South, code 6.

Thus, the chain code is:

$$\boxed{C=001234566}$$

Step 3: Normalisation with respect to starting point

To make the chain code independent of the starting point, circular shifts are compared and the smallest sequence is selected.

$$\boxed{C_{normal}=001234566}$$

The starting-point normalised code remains the same.

(b) Compute circular first difference and shape number.

Step 1: Calculate circular first difference

Formula:

$$d_i=(c_{i+1}-c_i)\mod 8$$

Given chain code:

$$C=0,0,1,2,3,4,5,6,6$$

Calculations:

Current	Next	First difference
0	0	0
0	1	1
1	2	1
2	3	1
3	4	1
4	5	1
5	6	1
6	6	0
6	0	2

Therefore:

$$\boxed{FD=011111102}$$

Step 2: Calculate shape number

The shape number is obtained by finding the smallest circular shift of the first-difference sequence.

$$FD=011111102$$

The smallest circular representation is:

$$\boxed{Shape\ Number=011111102}$$

(c) Calculate perimeter and enclosed area.

1. Perimeter

From the chain code:

- Horizontal/vertical movements = 6
- Diagonal movements = 3

Cardinal movement length = 1

Diagonal movement length = $\sqrt{2}$

Formula:

$$P=N_c+N_d\sqrt{2}$$

$$P=6+3\sqrt{2}$$

$$P=6+3(1.414)$$

$$\boxed{P=10.242\ units}$$

2. Enclosed Area

Using the shoelace formula:

$$A=\frac{1}{2}\left|\sum x_i y_{i+1}-\sum y_i x_{i+1}\right|$$

From the given coordinates:

$$\sum x_i y_{i+1}=25$$

$$\sum y_i x_{i+1}=10$$

$$A=\frac{1}{2}|25-10|$$

$$\boxed{A=7.5\ square\ units}$$

(d) Derive the new chain code after 90° counter-clockwise rotation and analyse the template

Step 1: Apply 90° counter-clockwise rotation

For 8-direction Freeman code, a 90° counter-clockwise rotation adds 2 to each direction code modulo 8.

$$C'= (C+2)\mod8$$

Original chain code:

$$C=0,0,1,2,3,4,5,6,6$$

New chain code:

$$\boxed{C'=2,2,3,4,5,6,7,0,0}$$

Step 2: Calculate new first difference

$$ FD'=(c'_{i+1}-c'_i)\mod8 $$

$$ \boxed{FD'=011111102} $$

The first difference remains unchanged because all direction codes are increased by the same amount.

Step 3: Which descriptor should be stored?

- Original chain code changes when the part rotates.
- First difference is rotation invariant.
- Shape number is also independent of the starting point.
- Therefore, the normalised first difference (shape number) should be stored as the inspection template.

Conclusion:

The Freeman chain code represents the boundary of the machined part. The first difference provides rotation invariance, while the shape number provides both starting-point and rotation invariance. Hence, it is suitable for automated shape inspection.